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PROSPERO
International prospective register of systematic reviews

Cross-sectional associations between serum cholesterol and calcium.

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REVIEW TITLE AND BASIC DETAILS

Review title

Cross-sectional associations between serum cholesterol and calcium.

Review objectives

Define the strength and consistency of the associations between serum calcium and serum cholesterol, including total, HDL and LDL-cholesterol.

Keywords

Osteoporosis; Nutrition; Total cholesterol; Cholesterol; Cardiovascular disease; Calcium; Serum Calcium; Demineralization; Bone loss

SEARCHING AND SCREENING

Searches

We will search MEDLINE (via PubMed), EMBASE, Web of Science, and the Cochrane Library from inception to August 2025. We will also search ClinicalTrials.gov and WHO ICTRP for unpublished or ongoing studies, and review the reference lists of included articles.

Searches will be limited to human studies but not restricted by publication date. We will initially include studies in all languages; articles not available in English will be translated if feasible.

Eligible study designs will include observational human studies (cross-sectional, cohort, and case-control) and randomized controlled trials reporting blood calcium and cholesterol measures. Animal studies, case reports, editorials, and narrative reviews will be excluded.

In addition to structured database searches, we will perform backward and forward citation chasing and use semantic/AI-based search tools (*e.g.* PubMed "similar articles", Semantic Scholar) to

identify potentially relevant studies not retrieved by keyword searches. These records will be screened with the same eligibility criteria, and their yield reported separately.

Study design

Both randomized and nonrandomized study types will be included.

Included

Eligible studies will include randomized controlled trials, prospective cohort studies, retrospective cohort studies, and case-control studies that report clinical associations between calcium and cholesterol (or subfractions of cholesterol) in blood.

Excluded

Case reports, commentaries, reviews, editorials, and animal studies will be excluded.

Link to search strategy

A full search strategy has been uploaded to PROSPERO. The PDF may be accessed through this link https://www.crd.york.ac.uk/PROSPEROFILES/4658ebb70947e010f159448d547d52f4.pdf.

ELIGIBILITY CRITERIA

Condition or domain being studied

Serum Calcium; Total Cholesterol Measurement; Calcium; Cholesterol; Low Density Lipoprotein Cholesterol Measurement; High Density Lipoprotein Cholesterol Measurement; Triglycerides Measurement

Population

Included

- Studies enrolling human participants of any sex and age
- Individuals with measured blood calcium and cholesterol levels
- Healthy volunteers, patients with established chronic disease, or specific clinical populations (e.g., cardiovascular, metabolic conditions), where blood measurements are taken for research or routine care
- Studies reporting clearly defined measurement methods for calcium and cholesterol concentrations in blood

Excluded

- Animal studies
- Studies without measured blood calcium or cholesterol data
- Studies focused exclusively on children (if relevant), or only on patients with rare genetic conditions not representative of clinical populations

Intervention(s) or exposure(s)

Included

Calcium; Total Cholesterol Measurement; Low Density Lipoprotein Cholesterol Measurement; High Density Lipoprotein Cholesterol Measurement; Triglycerides Measurement

Excluded
Atererial or aortal calcification or calcium imaging

Comparator(s) or control(s)
This review does not have any comparators

Context
Eligible studies will be those conducted in clinical or community settings involving human participants with documented blood calcium and cholesterol measurements. Studies from hospital inpatient, outpatient clinic, and population-based cohort settings will be included. Studies conducted exclusively in animal models or experimental laboratories will be excluded. Studies must report blood biochemical measurements obtained through standardized clinical assays or validated laboratory methods.

OUTCOMES TO BE ANALYSED

Main outcomes
The strength of the association between serum calcium and serum cholesterol measurements. Our primary focus will be on the correlation coefficient and beta coefficients from these associations. We will use Fisher-*z* transformed r's for meta-analysis and back-transform these for presentation and publication.

Additional outcomes
We will evaluate LDL-cholesterol, HDL-cholesterol and circulating triglycerides and their association with serum calcium as secondary outcomes. We will evaluate sex-specific and where possible associations in different populations and age groups as sensitivity analyses.

DATA COLLECTION PROCESS

Data extraction (selection and coding)
All records retrieved from database and supplementary searches will be imported into a reference management program (e.g. EndNote/Zotero) and de-duplicated. Screening will be performed in two stages by two independent reviewers:

• Title/abstract screening against the predefined eligibility criteria to exclude clearly irrelevant articles.

• Full-text review of potentially eligible articles to confirm inclusion and extract data.

Disagreements at either stage will be resolved through discussion and, if necessary, consultation with a third reviewer, and noted in a tracking document. Reasons for exclusion at full-text will be documented and all considered manuscripts will be reported as a supplementary file. Separate screening logs will be maintained for observational studies and randomized controlled trials. The entire process will be reported in a PRISMA flow diagram.

Risk of bias (quality) assessment

Study characteristics and risk of bias. For each included study we will extract:

- Bibliographic information (authors, year, journal, country).
- Study design (cross-sectional, cohort, case–control, randomized trial).
- Population details (sample size, age, sex distribution, pre/post-menopausal status, disease status such as primary hyperparathyroidism).
- Exposure/treatment (serum or plasma calcium measures; calcium/vitamin D supplementation; statin use).
- Outcome (type of cholesterol measure, units, assay method).
- Effect estimates (correlation coefficients r or effect sizes, adjustments performed, covariates).

For randomized controlled trials we will use the Cochrane RoB 2 tool; for observational association studies we will use the Newcastle–Ottawa Scale (NOS). If nonrandomized intervention studies are included, we will assess them with ROBINS-I.

Two reviewers will perform the bias assessments independently, with disagreements resolved by discussion or a third reviewer. Summary tables and plots of risk-of-bias judgments will be produced in R.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

We will extract correlation coefficients from primary research papers (separating RCTs from observational data), and when needed recalculate them from beta-coefficients and standard errors. Since correlation coefficients are not typically normally distributed, we will apply Fisher's z transformation, perform the meta-analyse on that scale, and then back-transform to r for presentation. Analyses will be performed in R using the meta package both by random- and fixed-effects models, as well as by Bayesian hierarchical models with a priors informed by our previous murine studies. Heterogeneity will be evaluated by I2, tau statistics and by visual analysis of forest plots, with statistical significance of heterogeneity reported in the manuscript. Subgroup analyses will be performed for sex, age groups and pre-/post-menopausal populations if sufficient data exists.

Analysis of subgroups or subsets

If sufficient data is available in underlying papers, we will perform subgroup analyses on the following populations

- Sex (male versus female participants)
- Age, including pre-/post-menopause status
- Treatment mode (statins, calcium or vitamin D supplementation, versus those not receivingthe , those treatments)
- Disease status (primary hyperparathyroidism or not)

For each factor we will either

- Extract stratum-specific effect sizes and pool them separately

- Conduct random-effects meta-regression when continuous or categorical covariates are reported across studies.

Subgroup analyses will be performed only when at least a minimum number of studies (i.*e.* ≥3 per subgroup) are available.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members

Dr Dave Bridges (review guarantor and contact) University of Michigan. United States of America.

No conflict of interest declared.

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No conflict of interest declared.

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Funding source

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NIAMS P30 AR069620

Additional non-commercial funding information
Pilot and Feasibiity Grant

Additional information about funding
Pilot and Feasibility award as part of AR069620

TIMELINE OF THE REVIEW

Review timeline

Start date: 14 February 2025. End date: 30 August 2026.

Date of first submission to PROSPERO

14 September 2025

Date of registration in PROSPERO

14 September 2025

CURRENT REVIEW STAGE

Publication of review results

Results of the review will be published in English.

Stage of the review at this submission

Review stage	Started	Completed
Pilot work	✓	
Formal searching/study identification		
Screening search results against inclusion criteria		
Data extraction or receipt of IPD		
Risk of bias/quality assessment		
Data synthesis		

Review status
The review is currently planned or ongoing.

ADDITIONAL INFORMATION

PROSPERO version history

- Version 1.0, published 14 Sep 2025

Review conflict of interest
Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

Country
United States of America

Other registration details
Extracted data will be deposited at the SRDR: Systematic Review Data Repository (link TBD) and on our group's GitHub repository for this project (https://github.com/BridgesLab/PrecisionNutrition) which in turn will also be archived permanently on Zenodo.

Medical Subject Headings
Calcium, Dietary; Cholesterol, LDL; Cross-Sectional Studies; Humans; Triglycerides; Calcium; Cholesterol; Cholesterol, HDL; Bone Diseases, Metabolic

Details of any existing review of the same topic by the same authors
No

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